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Markov Chain Convergence in Kernel-Induced Geometry

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Abstract

We study the convergence of Markov chains in the geometry induced by Hilbertian embeddings associated with positive definite kernels, also called Maximum Mean Discrepancy (MMD). Under a contraction assumption in the kernel-induced geometry, and a standard spectral gap assumption, we establish non-asymptotic, high-probability bounds for the convergence of the empirical occupation measure to the invariant measure. The obtained bounds separate the roles of the two conditions: the spectral gap determines the effective sample size in the leading term, while the kernel contraction enters only through a linear correction. As an application, we study one-step prediction of dynamical systems from a single trajectory, deriving fast learning rates for a nonlinear least-squares estimator. The rates highlight the interplay between mixing properties in the kernel-induced geometry and the classical source and capacity conditions used in the analysis of least-squares.

1 Introduction

Let $(X_t)_{t\geq 0}$ be a Markov chain on $\mathcal{X}$ with transition kernel P. Its *empirical occupation measure* up to time T is

$$\widehat{\mu}_T := \frac{1}{T}\sum_{t=0}^{T-1} \delta_{X_t}. \tag{1}$$

This random probability measure records the proportion of time spent by the trajectory in each measurable set. An important question in the analysis of Markov chains (and, more broadly, of dynamical systems) is whether there exists a probability measure π such that $\widehat{\mu}_T \to \pi$ as $T \to \infty$. Such a limit, when it exists, describes the statistical behavior of trajectories over long times (Eckmann and Ruelle, 1985; Young, 2002). For example, if $(X_t)_{t\geq 0}$ are independent random variables on a separable metric space $\mathcal{X}$ with common law π, the empirical law of large numbers gives, almost surely,

$$\widehat{\mu}_T \rightharpoonup \pi, \qquad \text{or, equivalently,} \qquad \int f\, d\widehat{\mu}_T = \frac{1}{T}\sum_{t=0}^{T-1} f(X_t) \to \int f\, d\pi \tag{2}$$

for every bounded continuous test function f. For general Markov chains, however, convergence to a deterministic limiting measure is not automatic. For classical conditions ensuring almost-sure convergence of $\widehat{\mu}_T$ as in (2), see, e.g., Meyn and Tweedie (2012).

To quantify the discrepancy between $\widehat{\mu}_T$ and π, we need to choose a distance on the space of probability measures. A flexible framework is provided by *integral probability metrics* (IPMs),

$$d_{\mathcal{F}}(\mu,\nu) := \sup_{f\in\mathcal{F}} \left| \int f\, d\mu - \int f\, d\nu \right|, \tag{3}$$

where $\mathscr{F}$ is a class of measurable test functions integrable with respect to the measures under consideration. The choice of $\mathscr{F}$ determines which differences between probability measures are detected, and hence the notion of convergence being studied. For example, the class of measurable functions satisfying $\|f\|_\infty \le 1$ gives the total variation norm, while the class of 1-Lipschitz functions gives the Wasserstein distance W_1 on probability measures with finite first moments. Further examples and general properties of IPMs can be found in Müller (1997) and Rachev et al. (2013).

Our aim is to obtain non-asymptotic bounds on $d_{\mathscr{F}}(\widehat{\mu}_T, \pi)$, for suitable choices of $\mathscr{F}$, with explicit dependence on the trajectory length T. Such bounds are useful in sampling, where they quantify the error incurred when approximating a target distribution by a finite trajectory of a Markov chain (Roberts and Rosenthal, 2004; Joulin and Ollivier, 2010). Related questions also arise in the analysis of stochastic optimization algorithms through their interpretation as Markov chains (Dieuleveut et al., 2020; Yu et al., 2021).

Non-asymptotic bounds have been studied in several probability metrics. For uniformly ergodic Markov chains on countable state spaces, bounds in total variation follow from the concentration and moment inequalities of Paulin (2015). Bounds in W_1 under suitable contractivity assumptions have been obtained by Boissard (2011); Kloeckner (2020); Riekert (2022). For W_p, Fournier and Guillin (2015) establish moment bounds for empirical measures of independent samples and Markov chains. Related bounds for occupation measures of continuous-time ergodic Markov processes are obtained by Schilling et al. (2025) under Wasserstein contractivity or functional inequalities, together with moment assumptions on the limiting measure.

In this work, we consider IPMs obtained from reproducing kernel Hilbert space (RKHS), known as *maximum mean discrepancies* (MMD). Let $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ be a measurable positive-definite kernel with separable RKHS $\mathscr{H}$, and let $\phi(x) := k(x, \cdot)$ be its canonical feature map. Taking $\mathscr{F}$ to be the unit ball of $\mathscr{H}$ in (3), Hilbert-space duality gives

$$\mathrm{MMD}_\phi(\mu, \nu) = \left\| \int \phi(x)\, \mu(dx) - \int \phi(x)\, \nu(dx) \right\|_{\mathscr{H}}, \tag{4}$$

whenever the two Bochner integrals exist; see, e.g., Smola et al. (2007); Gretton et al. (2012). The same construction applies to any measurable feature map $\phi : \mathcal{X} \to \mathscr{H}$ into a separable Hilbert space, subject to the same integrability requirement. The resulting discrepancy is the MMD associated with the kernel $k(x, y) = \langle \phi(x), \phi(y) \rangle_{\mathscr{H}}$. We develop this feature-map formulation in Section 3.

When the feature map is uniformly bounded, that is, $\sup_{x \in \mathcal{X}} \|\phi(x)\|_{\mathscr{H}} < \infty$, the discrepancy is well defined for all probability measures, including empirical occupation measures. For independent observations with common law π, $\mathrm{MMD}_\phi(\widehat{\mu}_T, \pi)$ admits bounds of order $T^{-1/2}$, both in expectation and at each fixed confidence level (Gretton et al., 2012). For dependent observations, Chérief-Abdellatif and Alquier (2022) obtain high-probability bounds under weak-dependence assumptions in their analysis of minimum-MMD estimators. More recently, Wolfer and Alquier (2025) derive such bounds for stationary φ- and β-mixing sequences.

Contributions. Our first contribution is a non-asymptotic high-probability bound for empirical occupation measures of Markov chains on a general state space. Assume that the chain is stationary, has an absolute $L^2(\pi)$ spectral gap $1 - \rho > 0$, and satisfies the following uniform geometric relaxation condition:

$$\sup_{x \in \mathcal{X}} \mathrm{MMD}_\phi(P^n(x, \cdot), \pi) \le C\theta^n, \qquad n \ge 0, \tag{5}$$

for some $C < \infty$ and $\theta \in [0, 1)$. Then, for every $T \ge 1$ and $\delta \in (0, 1)$, with probability at least $1 - \delta$,

$$\mathrm{MMD}_\phi(\widehat{\mu}_T, \pi) \lesssim \|\phi\|_\pi \sqrt{\frac{\tau_{\mathrm{rel}} \log(4/\delta)}{T}} + \frac{\tau_\theta \sqrt{\tau_{\mathrm{rel}}} \log(4/\delta)}{T}, \tag{6}$$

where $\tau_{\mathrm{rel}} := (1-\rho)^{-1}$, $\tau_\theta := C/(1-\theta)$, and $\|\phi\|_\pi^2 := \int \|\phi(x)\|_{\mathscr{H}}^2 \, \pi(dx)$; the implicit constant is universal. The leading term depends on the spectral gap and the second moment of the feature map, while the relaxation constants C and θ enter only through the T^{-1} term. We extend this bound to initial laws $\nu \ll \pi$ with $d\nu/d\pi \in L^p(\pi)$ for some $p > 1$, leaving the leading term unchanged; see Theorems 3.3 and 3.6. We also give sufficient conditions for (5).

Our second contribution concerns kernel ridge regression (KRR) for learning the one-step prediction function

$$f_\star(x) := \int x' \, P(x, dx') \tag{7}$$

of a real-valued Markov chain from a single trajectory. The relevant MMD geometry arises from the covariance fluctuations controlling estimator stability. Writing Σ and $\widehat{\Sigma}$ for the population and empirical covariance operators, and denoting $\Sigma_\lambda := \Sigma + \lambda I$, we have

$$\left\| \Sigma_\lambda^{-1/2} (\widehat{\Sigma} - \Sigma) \Sigma_\lambda^{-1/2} \right\|_{\mathscr{L}_2(\mathscr{H})} = \mathrm{MMD}_{\Phi_\lambda}(\widehat{\mu}_T, \pi), \tag{8}$$

where $\Phi_\lambda(x) := \Sigma_\lambda^{-1/2}[\phi(x) \otimes \phi(x)]\Sigma_\lambda^{-1/2}$. We obtain relaxation bounds in this λ-dependent geometry from contraction in a fixed Hilbert scale associated with the covariance-feature kernel $K(x,y) = k(x,y)^2$.

For stationary trajectories, under the spectral and Hilbert-scale assumptions, and classical source and capacity conditions of Theorem 4.4, we obtain

$$\mathscr{E}(\widehat{f}_{\lambda_T}, f_\star) = \left\| \widehat{f}_{\lambda_T} - f_\star \right\|_\pi^2 \lesssim T^{-\frac{2r}{2r+p}}, \qquad \lambda_T \asymp T^{-\frac{1}{2r+p}}. \tag{9}$$

For every $\delta \in (0,1)$, this bound holds with probability at least $1-\delta$ for each sufficiently large T, up to logarithmic factors in $1/\delta$. Here, $r \in [1/2, 1]$ is the source exponent and $p = 1/b \in (0,1)$ is the capacity exponent, characterized by $\mathcal{N}(\lambda) \lesssim \lambda^{-p}$. The rate matches the classical source–capacity exponent for i.i.d. KRR (Caponnetto and De Vito, 2007), with explicit dependence on the chain in the finite-sample bounds. By incorporating source and capacity information, this sharpens the $T^{-1/2}$ excess-risk rate of Kachaiev et al. (2026), however, under different assumptions. Our work considers only stationary initialization while relaxing uniform ergodicity in total variation to absolute spectral gap, and the Hilbert-scale contraction described above.

Structure of the paper. Section 2 collects background on Markov operators, martingales, integral probability metrics, and spectral gaps, together with preliminary results used throughout the paper. Section 3 introduces uniform MMD relaxation and presents the occupation measure bounds, followed by sufficient conditions and examples based on metric comparison and contraction in kernel-induced Hilbert scales. Section 4 applies these results to KRR and derives the learning guarantees. The proofs, together with the concentration inequalities they rely on, are collected in Appendix A; Appendix B contains the Fourier computations for the torus examples, including the verification of all assumptions of Theorem 4.4 for a noisy doubling map.

2 Background and preliminaries

Tools and notation. Let $(\mathcal{X}, d)$ be a Polish space endowed with Borel σ-algebra $\mathscr{B}(\mathcal{X})$. Let $\mathscr{P}(\mathcal{X})$ be a space of probability measures on $(\mathcal{X}, \mathscr{B}(\mathcal{X}))$. In the context where $\mathcal{X}$ is a normed space, $\mathscr{P}_2(\mathcal{X})$ denotes the subset of probability measures on $\mathcal{X}$ with finite second moment, i.e., $\int \|x\|^2 \, \mu(dx) < \infty$. Define $\mathrm{diam}(\mathcal{X}) := \sup_{x,y \in \mathcal{X}} d(x,y)$.

Let $(X_t)_{t \geq 0}$ be a Markov process taking values in $(\mathcal{X}, \mathscr{B}(\mathcal{X}))$, with associated Markov transition kernel $P : \mathcal{X} \times \mathscr{B}(\mathcal{X}) \to [0,1]$. Assume that P admits an invariant probability measure $\pi \in \mathscr{P}(\mathcal{X})$,

that is, $\pi P = \pi$, where, for $\mu \in \mathcal{P}(\mathcal{X})$, we write μP for the probability measure

$$(\mu P)(A) := \int P(x, A)\, \mu(dx), \qquad A \in \mathcal{B}(\mathcal{X}). \tag{10}$$

The process $(X_t)_{t\geq 0}$ is called *stationary* if $X_0 \sim \pi$. We also use the notation $P^n(x, \cdot) := \delta_x P^n$ for $n \geq 0$. Recall that for a fixed $T > 0$,

$$\widehat{\mu}_T := \frac{1}{T} \sum_{t=0}^{T-1} \delta_{X_t} \tag{11}$$

denotes empirical *occupation measure*. For a measure μ and a measurable function $f : \mathcal{X} \to \mathbb{R}$, let $\mu(f) := \int f \, d\mu$.

Let $L^2(\pi)$ denote the Hilbert space of square-integrable functions $f : \mathcal{X} \to \mathbb{R}$, endowed with the inner product $\langle f, g\rangle_\pi := \pi(fg)$. Denote the subspace of π-centered functions by $L^2_0(\pi) := \{f \in L^2(\pi) : \pi(f) = 0\}$. For a separable Hilbert space $\mathcal{Y}$, we let $L^2(\mathcal{X}, \mathcal{B}(\mathcal{X}), \pi; \mathcal{Y})$, abbreviated $L^2(\pi; \mathcal{Y})$, denote the Bochner space of strongly $\mathcal{B}(\mathcal{X})$-measurable and Bochner square-integrable maps $f : \mathcal{X} \to \mathcal{Y}$ with

$$\|f\|^2_\pi := \|f\|^2_{L^2(\pi;\mathcal{Y})} = \int \|f(x)\|^2_{\mathcal{Y}}\, \pi(dx) < \infty. \tag{12}$$

Denote by $\mathcal{L}(\mathcal{H}_1, \mathcal{H}_2)$ be the space of bounded linear operators between two separable Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$, and $\mathcal{L}(\mathcal{H}_1)$ when $\mathcal{H}_1 = \mathcal{H}_2$. For $A \in \mathcal{L}(\mathcal{H}_1)$, denote the operator norm by $|\!|\!| A |\!|\!|_{\mathcal{H}_1} := \|A\|_{\mathcal{H}_1 \to \mathcal{H}_1}$. Let $\mathcal{L}_2(\mathcal{H}_1, \mathcal{H}_2)$ be the space of Hilbert–Schmidt operators, and $\mathcal{L}_2(\mathcal{H}_1)$ when $\mathcal{H}_1 = \mathcal{H}_2$, equipped with the norm $\|A\|^2_{\mathcal{L}_2(\mathcal{H}_1,\mathcal{H}_2)} = \sum_i \|Ae_i\|^2_{\mathcal{H}_2}$, where $\{e_i\}_i$ is any orthonormal basis of $\mathcal{H}_1$. We denote by I_d the $d \times d$ identity matrix.

Martingales. Let $(\mathcal{F}_t)_{t\geq 0}$ be a filtration. A sequence of $\mathcal{H}$-valued random variables $(d_t)_{t\geq 1}$ is called a *martingale difference sequence* with respect to $(\mathcal{F}_t)_t$ if d_t is $\mathcal{F}_t$-measurable and for every $t \geq 1$, $\mathbb{E}[d_t \mid \mathcal{F}_{t-1}] = 0$. Setting $M_0 := 0$ and $M_T := \sum_{t=1}^T d_t$ define an $\mathcal{H}$-valued *martingale*. Its *predictable quadratic variation* is

$$\langle M\rangle_T := \sum_{t=1}^T \mathbb{E}_{t-1}\|d_t\|^2, \tag{13}$$

where $\mathbb{E}_j[\cdot] := \mathbb{E}[\cdot \mid \mathcal{F}_j]$. For background on martingales and martingale difference sequences, we refer to Hall and Heyde (1980); Williams (1991).

Integral probability metrics. Let $\mathcal{F}$ be a class of measurable real-valued functions on $\mathcal{X}$. For $\mu, \nu \in \mathcal{P}(\mathcal{X})$ such that the integrals below are well defined, the *integral probability metric (IPM)* generated by $\mathcal{F}$ is

$$d_{\mathcal{F}}(\mu, \nu) := \sup_{f\in\mathcal{F}} |\mu(f) - \nu(f)|\,. \tag{14}$$

The choice of $\mathcal{F}$ determines the geometry on the space of probability measures. In particular, for the class

$$\mathcal{F}_{\mathrm{TV}} := \{f : \mathcal{X} \to [0, 1] : f \text{ is measurable}\}\,, \tag{15}$$

the corresponding IPM is the total variation distance:

$$d_{\mathcal{F}_{\mathrm{TV}}}(\mu, \nu) = \sup_{A\in\mathcal{B}(\mathcal{X})} |\mu(A) - \nu(A)| = \|\mu - \nu\|_{\mathrm{TV}}. \tag{16}$$

Suppose that $(\mathcal{X}, d)$ is a bounded space, i.e., $\mathrm{diam}(\mathcal{X}) < \infty$. Fix $x_0 \in \mathcal{X}$ and consider the anchored Lipschitz class

$$\mathcal{F}_{\mathrm{Lip}(d)} := \{f : \mathcal{X} \to \mathbb{R} : |f(x) - f(y)| \leq d(x, y)\ \forall x, y \in \mathcal{X},\ f(x_0) = 0\}\,. \tag{17}$$